[1]:
```python
print("===LIST EXAMPLE ===")
fruits=["apple","banana","cherry","apple"]
print("Original List:",fruits)
```

```
===LIST EXAMPLE ===
Original List: ['apple', 'banana', 'cherry', 'apple']
```

[2]:
```python
print("===LIST EXAMPLE ===")
fruits=["apple","banana","cherry","apple"]
print("Original List:",fruits)#Add element
fruits.append("orange")
#Remove elemant
if"banana" in fruits:
    fruits.remove("banana")
print(" first fruit:",fruits[0])
```

```
===LIST EXAMPLE ===
Original List: ['apple', 'banana', 'cherry', 'apple']
 first fruit: apple
```

[3]:
```python
print("\n===TUPLE EXAMPLE===")
coordinates=(10.5,20.3,10.5)
print("Tuple:",coordinates)
print("First coordinate:",coordinates[0])
```

```
===TUPLE EXAMPLE===
Tuple: (10.5, 20.3, 10.5)
First coordinate: 10.5
```

[11]:
```python
print("\n===TUPLE EXAMPLE===")
coordinates=(10.5,20.3,10.5)
print("Tuple:",coordinates)
print("First coordinate:",coordinates[-1])
x,y,z=coordinates
print(f"unpacked values:x={x},y={y},z={z}")
```

```
===TUPLE EXAMPLE===
Tuple: (10.5, 20.3, 10.5)
First coordinate: 10.5
unpacked values:x=10.5,y=20.3,z=10.5
```

[5]:
```python
# Create a list of 5 numbers and print all elements.
numbers = [10, 20, 30, 40, 50]

# Print all elements
for num in numbers:
    print(num)
```

```
10
20
30
40
50
```

[7]:
```python
# Given list
numbers = [10, 20, 30, 40, 50]

# First element
print("First element:", numbers[0])

# Last element
print("Last element:", numbers[-1])
```

```
First element: 10
Last element: 50
```

[6]:
```python
# Initial list
numbers = [1, 2, 3, 4]

# Number to add
num_to_add = 100

# Validate that the value is an integer before adding
if isinstance(num_to_add, int):
    numbers.append(num_to_add)  # Adds to the end of the list
    print("Updated list:", numbers)
else:
    print("Error: Only integers can be added.")
```

```
Updated list: [1, 2, 3, 4, 100]
```

[8]:
```python
# Original list
numbers = [1, 2, 3, 4, 5]

try:
    # Remove the first occurrence of 3
    numbers.remove(3)
except ValueError:
    # This will run if 3 is not found in the list
    print("Element 3 not found in the list.")

print("Updated list:", numbers)
```

```
Updated list: [1, 2, 4, 5]
```

[9]:
```python
numbers = [5, 10, 15, 20]

total = 0
for num in numbers:
    total += num

print("Sum:", total)
```

```
Sum: 50
```

[10]:

```python
# Program to find the maximum and minimum in a list

def find_min_max(numbers):
    """
    Returns the minimum and maximum values from a list of numbers.
    Includes validation to ensure the list is not empty and contains only numbers.
    """
    if not numbers:
        raise ValueError("The list is empty.")
    if not all(isinstance(x, (int, float)) for x in numbers):
        raise TypeError("All elements in the list must be integers or floats.")

    # Using built-in functions for efficiency
    min_val = min(numbers)
    max_val = max(numbers)
    return min_val, max_val
```

```python
if __name__ == "__main__":
    data = [8, 3, 9, 1, 6]
    try:
        minimum, maximum = find_min_max(data)
        print(f"List: {data}")
        print(f"Minimum value: {minimum}")
        print(f"Maximum value: {maximum}")
    except (ValueError, TypeError) as e:
        print(f"Error: {e}")
```

```
List: [8, 3, 9, 1, 6]
Minimum value: 1
Maximum value: 9
```